

A Byte-Level Multilingual News Recommender for Microcontrollers

Abstract. Running a news recommender on the device itself keeps a reader's clicks private, but the usual models store a word-embedding table of tens to hundreds of megabytes per language, which no microcontroller's flash can hold. We avoid the table by reading a title as its raw UTF-8 bytes: a small char-CNN whose 256-value vocabulary is shared by every language. Around this encoder we search three families of architecture (unconstrained, budget-constrained and binary-aware) and evaluate each at full precision and with inner-layer weights at eight bits and at one bit, measuring ranking quality on MIND against estimated flash, RAM and energy. With eight-bit weights the constrained model scores 0.610 AUC at an estimated 204KB and the unconstrained one 0.647 at 790 KB. For reference, NRMS with GloVe vectors scores 0.664 over three seeds, a reduced NRMS with a random 256-dimensional table 0.604, and a per-user category histogram with no network 0.598. At the constrained architecture, starting from weights distilled from a frozen multilingual sentence-transformer adds 0.011 AUC (0.629 against 0.618), and training on clicks shown in a random xMIND language raises the mean over fourteen translations from 0.557 to 0.589 at a cost of 0.020 in English. A naive one-bit model stays near chance (0.521); ReActNet blocks started from the distilled weights bring it to 0.572 at 194KB. Seeded reruns differ from the matrix run by up to 0.02 AUC, so smaller gaps are not resolved. Exported as an integer-only file, the distilled constrained encoder scores 0.622 (0.628 at full precision) and runs in 31.2 ms per title on an STM32H7B3I-DK and 450 ms on a NUCLEO-F401RE. **[TODO: energy on the boards and timing on an Android phone]**. Code and artefacts are released.

Keywords: Edge AI · Neural Architecture Search · Binarized Neural Networks · Multilingual News Recommendation · TinyML · Knowledge Distillation

1 Introduction

Running personalisation on the phone or sensor node rather than the cloud is attractive for privacy: the record of what a user reads never leaves the device. Two difficulties come with that setting. A model that starts from nothing for each new user recommends poorly until it has seen enough clicks, and the standard neural recommenders (NRMS [21], NAML [20] and their relatives) represent a title with a word-embedding matrix. That matrix is large: GloVe-300d over 30–100k words is already 37–120 MB, one table per language, and none of it fits a microcontroller's few megabytes of flash.

We study both problems offline, on MIND [23] and its translated extension xMIND [6], and then shrink the result for the device. The idea is to drop words and read a title as its UTF-8 bytes. A byte takes 256 values, so the vocabulary is a single 256-row table shared by every language, and one small convolutional encoder reads Chinese, Georgian and English alike. A frozen multilingual teacher [17] supplies distillation targets that place a translated title near its English original; we use them to score candidate architectures and to initialise the encoder.

We contribute three things. A byte-level content encoder removes the embedding-table memory wall: its vocabulary is one 256-row table for every language. Around it we benchmark three architecture searches (unconstrained, budget-constrained and binary-aware) at three weight precisions, reporting accuracy alongside estimated flash, RAM and energy. Finally we bring the one-bit case from near chance to 0.572 AUC with ReActNet activations and Bi-Real shortcuts started from distilled weights. Character-level inputs have been used for click prediction [3], and vocabulary-free projections give text classifiers of about 200KB [7]. A search of arXiv and IEEE Xplore (October 2026) returned no news recommender that reads raw bytes and none evaluated on a microcontroller; the nearest on-device NAS and binary work is on vision [9, 10, 15].

2 Related Work

Neural news recommendation on MIND is well charted. NRMS [21], NAML [20], LSTUR, NPA and DKN encode titles and users with attention or knowledge graphs; their published test AUC on the large split runs from 0.65 (DKN) to 0.68 (NRMS) [23]. Language-model recommenders (PLM-NR, Fastformer, UNBERT) reach 0.70–0.73 [22] but wrap a BERT-sized encoder of hundreds of megabytes, out of reach on a microcontroller; we keep NRMS as the reference that is closest in size. The tools we borrow are vocabulary-free character and byte models [24, 4, 2], cross-lingual distillation [17], memory-budgeted NAS [9], and the binary-network line of XNOR-Net [16], Bi-Real [12] and ReActNet [11]; what is new is putting them together for on-device multilingual news.

3 Method

We treat recommendation as impression-level ranking: score candidate titles against the titles a user has clicked, embedding both with the same encoder (Fig. 1). The encoder reads a title’s bytes through a 256-entry embedding, a few depthwise-separable one-dimensional convolutions, a masked average over positions, and a linear projection. An additive-attention encoder summarises the user’s history, and the score is the dot product of the two vectors, trained with one positive and four negatives redrawn at every step (AdamW). A frozen `paraphrase-multilingual-MiniLM-L12-v2` [19] supplies a 384-dimensional target for each English title, and because the xMIND titles are translations of those

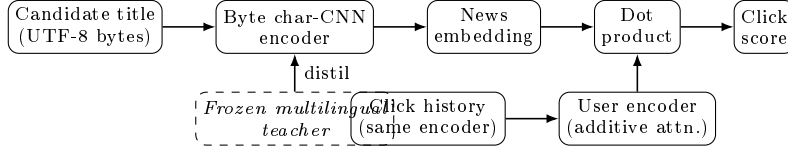


Fig. 1. The recommender: a byte char-CNN embeds candidate and history titles; an additive-attention encoder pools the history and a dot product scores the candidate. A frozen multilingual teacher provides distillation targets. Only the char-CNN is searched and quantised.

same articles the distillation loss maps every translation to that one target; a Matryoshka [8] term keeps the leading coordinates informative on their own. Distillation has two uses here: it is the fitness of the architecture search, and it initialises the encoder in the ablation and in the one-bit model of Sect. 5. The nine models of Table 1 are trained on clicks from a random initialisation.

The search has three arms: one unconstrained, one held to an on-device budget with eight-bit weights, and one held to the same budget with one-bit weights. The space holds 120 architectures (six channel counts, five depths, four output widths). Each arm runs an evolutionary search over it, scores a candidate by a short distillation, and rejects candidates over budget; precision is then the second axis. Quantisation is simulated on the weights with a straight-through estimator, per-channel for eight bits and sign-times-scale for one. Following XNOR-Net, the byte embedding and the outermost layers stay in full precision, and so do all activations, so the quantised rows cover 66%, 31% and 24% of the parameters of the three architectures. For the binary arm alone we add the ReActNet activations and Bi-Real shortcuts, binarise the block inputs and start from the distilled weights, which is what makes one-bit training converge to something useful. Sizes are parameter counts at each layer’s precision and multiply-accumulates are counted per layer. The energy column multiplies them by per-operation figures: 4.6 and 0.23 pJ for a 32-bit float and an eight-bit integer multiply-accumulate at 45 nm [5], and one sixty-fourth of the former for a one-bit operation, the operation-count convention of binary networks [16, 12]. We read it only as a relative ordering, since it ignores the memory traffic that often dominates on a real chip.

4 Experimental Setup

We use MINDsmall [23]: fifty thousand users, about 157k training and 73k development impressions over 51k articles, together with the fourteen NLLB translations of xMIND [6, 14]. xMIND provides only translated titles, keyed to the same article ids, so we reuse the English behaviour logs and swap the text. Every recommender we train, and the NRMS baseline, sees only English clicks, and the other languages appear only at evaluation. The multilingual numbers are therefore a test of cross-lingual transfer in the content encoder, not of native-language personalisation. We stay on the small split rather than the million-user MIND-

Table 1. Architecture $\times$ weight precision on MINDsmall dev, single runs. **Bold:** best per group; †: precision the arm was searched at. Arm = the search, precision = simulated weight quantisation, so off-diagonal cells are ablations. Size and energy are estimates (Sect. 3).

Arm	Prec.	AUC	MRR	nDCG@10	Size (KB)	Energy (μ J)
NAS	FP32 [†]	0.633	0.326	0.375	1566.8	166.3
NAS	INT8	0.647	0.338	0.387	789.8	17.91
NAS	Binary	0.588	0.297	0.342	563.1	12.54
Micro-NAS	FP32	0.605	0.307	0.352	266.8	15.15
Micro-NAS	INT8 [†]	0.610	0.314	0.360	203.9	3.16
Micro-NAS	Binary	0.521	0.251	0.294	185.6	2.72
Bin. μ NAS	FP32	0.593	0.290	0.336	311.4	14.98
Bin. μ NAS	INT8	0.601	0.295	0.344	255.7	4.35
Bin. μ NAS	Binary [†]	0.546	0.273	0.317	239.5	3.96

large because the full study — three arms, three precisions, the search and the fifteen-language evaluation — already takes 8.2 hours on one laptop; the small split has no public test set, so we evaluate on dev with the number of epochs fixed in advance. The search runs a population of thirty over fifteen generations, scoring each candidate by a two-epoch distillation on twenty thousand titles. The budgets of the constrained arms are 256 KB of flash and 128 KB of RAM, about eight times tighter than an STM32H7 holds. A bound of 2 M multiply-accumulates was also set, but the counter of that run skipped the quantised layers, so both constrained winners (3.29 M and 3.26 M) exceed it; **[TODO: re-run the constrained search with the corrected count]**. Everything runs on one laptop: an Intel Core Ultra 9 275HX with an eight-gigabyte RTX 5070 Laptop GPU and 32 GB of memory, PyTorch 2.12 on CUDA 13. The targets are STM32 microcontrollers and an Android phone. Board latency, flash and RAM come from the ST Edge AI Developer Cloud (ST Edge AI Core 4.0.1) on an STM32H7B3I-DK (Cortex-M7, 280 MHz) and a NUCLEO-F401RE (Cortex-M4, 84 MHz); **[TODO: phone timing and board energy]**. Negative sampling and the search are seeded; network initialisation in the matrix run was not, which the seeded reruns of Sect. 5 quantify. The downloaded data is pinned by checksum, and the code, notebook, result files and a dependency lock are released.¹ Table 1 comes from the June run, which drew the negatives once; the experiments of Table 2 use the loop of Sect. 3, which redraws them at every step, and report the mean and standard deviation over three seeds.

5 Results

Table 1 collects the nine combinations (the three arms found the architectures 256-4-384, 64-5-384 and 96-2-384, with 36.1, 3.29 and 3.26 million multiply-accumulates per title). Two patterns emerge. Moving the inner weights from full to eight-bit precision changes AUC by at most 0.014, inside the run-to-run variation reported below, while the energy estimate falls 3.4 to 9.3 times and

¹ <https://github.com/SepehrMohammady/MIND-Edge-Recommender>

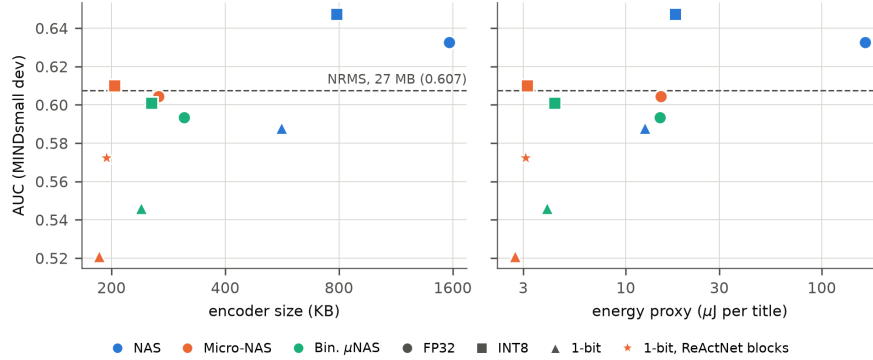


Fig. 2. Accuracy against estimated encoder size (left) and against the energy estimate (right), per arm and weight precision; dashed line: reduced NRMS with a 27 MB word table.

Table 2. Reference points and the reference byte-CNN (64-5-384, full precision) on MINDsmall dev. Mean $\pm$ standard deviation over 3 seeds; the last column is the mean AUC over the fourteen xMIND translations; “mixed clicks” = each training instance shown in a random xMIND language. Histogram scorers use MIND’s editorial labels. Params = news encoder.

Scorer	Params	AUC	MRR	nDCG@10	14-lang. AUC
Popularity	–	0.525	0.255	0.299	–
Category histogram	–	0.598	0.307	0.354	–
Subcategory histogram	–	0.617	0.341	0.384	–
NRMS, reduced (random 256-d table)	7.08 M	0.604 ± 0.010	0.322 ± 0.009	0.364 ± 0.009	–
NRMS, GloVe reference config.	11.29 M	0.664 ± 0.003	0.369 ± 0.002	0.415 ± 0.002	–
Frozen teacher + user encoder	–	0.589 ± 0.000	0.319 ± 0.000	0.360 ± 0.000	0.575 ± 0.000
+ trainable linear map	–	0.618 ± 0.002	0.327 ± 0.001	0.374 ± 0.001	0.590 ± 0.002
Byte-CNN, random start, EN clicks	68 k	0.618 ± 0.002	0.329 ± 0.007	0.371 ± 0.004	0.528 ± 0.002
Byte-CNN, random start, mixed clicks	68 k	0.584 ± 0.005	0.308 ± 0.013	0.348 ± 0.009	0.557 ± 0.001
Byte-CNN, distilled, frozen	68 k	0.573 ± 0.004	0.297 ± 0.002	0.340 ± 0.003	0.572 ± 0.001
Byte-CNN, distilled start, EN clicks	68 k	0.629 ± 0.003	0.338 ± 0.005	0.383 ± 0.003	0.557 ± 0.002
Byte-CNN, distilled start, mixed clicks	68 k	0.609 ± 0.008	0.312 ± 0.004	0.360 ± 0.006	0.589 ± 0.005

the size 1.2 to 2 times. One bit is different: it loses five to nine AUC points, and because the embedding and outer layers stay in full precision its size advantage over eight bits is small at this scale. The RAM estimate, the input and output buffer of the widest layer at one byte per activation, is 16 KB for the constrained model and 64 KB for the unconstrained one. On the laptop a title takes 0.10–0.34 ms on CPU (onnxruntime) and under 0.7 ms on GPU for the three full-precision architectures.

Table 2 places these numbers. Our NRMS reproduction in the reduced configuration of the June run (a randomly initialised 256-dimensional word table; 92% of its 7.08 million parameters are the table, 27 MB) averages 0.604 over three seeds with a spread of 0.010, which covers the 0.607 of the matrix run. In the reference configuration shipped with MIND (GloVe-initialised 300-dimensional table, twenty heads) NRMS reaches 0.664, in line with the 0.656 reported for this

split [13]. Two scorers with no network set a floor: popularity gives 0.525, and a histogram of the reader’s clicks over MIND’s category labels 0.598, or 0.617 with the finer subcategory labels. The reference byte-CNN trained on English clicks scores 0.618 from a random start and 0.629 from the distilled weights, so distillation is worth 0.011 here, less than the 0.033 of the June ablation, which drew negatives once and trained for ten epochs. The gap to GloVe NRMS is 0.036 with 165 times fewer parameters, and the distilled encoder is 0.011 above the subcategory histogram, which has no network at all. Feeds outside MIND carry no such editorial labels, which is the case a content encoder is for.

Cross-lingual transfer depends on how the encoder is trained (last column of Table 2). Trained from a random start on English clicks it reads the fourteen translated dev sets at 0.528 on average, close to chance; the June matrix models are of this kind. The distilled student alone, frozen under a trained user encoder, scores 0.572 across languages against 0.573 in English, so the distillation does carry the teacher’s language alignment (the teacher itself: 0.575 against 0.589). Fine-tuning the distilled encoder on English clicks raises English to 0.629 but leaves the translations at 0.557; showing the same clicks in a random xMIND language instead keeps 0.589 across languages, from 0.538 (Georgian) to 0.609 (Finnish), for 0.609 in English. A trainable linear map on the frozen teacher reaches 0.590 across languages with the whole teacher behind it; the 68k-parameter byte-CNN comes within 0.001 of that figure.

Two further points qualify the tables. Three seeded reruns of the eight-bit constrained model score 0.630, 0.622 and 0.621; the matrix run, whose initialisation was not seeded, scored 0.610, so we read gaps below about 0.02 in Table 1 as unresolved. For the 3% of impressions whose reader has no history, a history-based score ties every candidate (0.500 by construction); popularity gives 0.529 there and the population’s category distribution 0.533. By history length, the distilled encoder trained on English clicks scores 0.615 for readers with one to five clicks, 0.633 for six to twenty and 0.638 beyond, against 0.622, 0.664 and 0.688 for GloVe NRMS: the gap widens with history.

Binarising the inner weights the naive way leaves the constrained model near chance (0.521). Replacing its blocks with full binary convolutions, ReActNet activations and Bi-Real shortcuts, and starting from distilled weights, gives 0.572 at an estimated 194KB, 0.035 below the reduced NRMS. The 0.05 gain exceeds the run-to-run variation; three seeded runs of this model on a reduced training budget have a standard deviation of 0.012. On the accuracy-versus-footprint view (Fig. 2) eight-bit Micro-NAS (0.610, 204KB, 3.2 μ J) has the lowest cost at the accuracy of the reduced NRMS, while unconstrained eight-bit leads on accuracy (0.647, 790KB).

For the boards, the trained constrained encoder (distilled start, English clicks) was exported to ONNX without its byte-table lookup, which the caller performs, and quantised to an 8-bit QDQ model with ONNX Runtime (per-channel 8-bit weights, 8-bit activations, 512 training titles for calibration). The integer file scores 0.622 AUC against 0.628 for the same weights at full precision. On the STM32H7B3I-DK it runs in 31.2 ms per title, 2.5 times faster than at

full precision (77.0 ms), with 53.2 KB of weights, 77.7 KB of flash including the runtime and 49.4 KB of RAM, plus the 64.3 KB byte table; on the NUCLEO-F401RE it takes 450 ms, no faster than full precision there (444 ms). The board’s own check of the generated code against the reference model gave zero error.

6 Limitations

The main gap is hardware. We timed the matrix models on the laptop and estimated size, RAM and energy from parameter and operation counts; only the deployed encoders ran on the boards, and nothing ran on a phone. In the matrix the quantisation is simulated on weights only: activations, the byte embedding and the outer layers stay in floating point, so the eight-bit and one-bit rows may overstate accuracy and their sizes are not those of an integer-only file; for the constrained encoder the integer-only export costs 0.006 AUC, and the matrix is to be rerun with integer exports. Eight-bit inference runs on STM32 boards through quantised ONNX models and ST Edge AI, as measured above. The one-bit path needs a binary runtime such as CBin-NN [18] or Larq Compute Engine [1], and the Cortex-M7 has no popcount instruction, so we make no speed claim for binary and treat it only as a way to save space. The energy figure ignores SRAM and memory traffic, which can dominate on a Cortex-M. The constrained search enforced its flash and RAM budgets but not the operation budget, and the precision matrix predates the training loop of Table 2, so it will be rerun with it. Even after mixed-language training the translations stay 0.020 below English and 0.039 below the English-only encoder. Finally, a byte sequence is about five times longer than the word sequence, trading flash for computation, and all of this is on the small MIND split.

7 Ethics and Conclusion

MIND and xMIND are used under their non-commercial research licences; the work touches no personal data beyond the public logs, and the translations are machine-generated. Replacing the word-embedding table with a byte-level encoder removes the largest item of a news recommender’s storage and gives every language the same 256-row vocabulary. With eight-bit weights the constrained encoder reaches 0.610 AUC at an estimated 204 KB, and as an integer-only file it runs in 31.2 ms per title on an STM32H7B3I-DK; the one-bit model, started from distilled weights, reaches 0.572 at 194 KB; trained on clicks shown in a random language, the distilled byte-CNN reads fourteen languages at 0.589 AUC on average. What is still missing: **[TODO: board energy, phone timing and a one-bit runtime]** — that is where we take it next.

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